Experiment B: Rank 16 at the Fixed Two-Epoch SFT Recipe

Status: Completed; evaluated on 12 October 2026. Rank 8 remains the reference.

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Context

The anchored two-epoch SFT reference remains our reference for balanced ACT / CLARIFY / REFUSE routing. More SFT epochs and the DPO trials have not established a better balanced model. This experiment tests adapter capacity without changing the data or training objective.

What Changes

Setting	A: reference	B: candidate
LoRA rank	8	16
LoRA alpha	16	32
Alpha / rank	2	2
Target modules	q_proj, v_proj	q_proj, v_proj

Rank and alpha change together to retain the existing LoRA scaling ratio. This doubles the trainable adapter parameter count for the same target modules. It does not test adding k_proj/o_proj, lowering the learning rate or adding data.

Fixed Recipe

Setting	Both runs
Initialization	Fresh Qwen/Qwen2.5-1.5B-Instruct, not a continued adapter
Dataset	xlam-baymax-target-compatible-v14-act200-anchored-val200
Dataset revision	329fab4d8f1604a4085d3ec50e05881e7c069717bb1428db7e592743bb07f03c
Prompt SHA256	a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468
Raw caps	1,000 training / 200 validation rows
Actual sequence limit	1,024 with the command below

Setting	Both runs
Actual batch / accumulation	1 / 16 with --local-low-vram
Learning rate / epochs	0.0002 / 2
Dropout / seed	0.05 / 1
Scheduler / warmup	Cosine / 0.03

The JSON retains the reference's declared 2,048 context and batch settings; the same command-line overrides reproduce its actual low-VRAM recipe. The reference retained 974 training and 195 validation rows after token filtering. The candidate's final usable-row counts are not embedded in its evaluation JSON; the reference counts must not be presented as newly measured candidate counts.

Run

From the repository root in PowerShell:

```
$config =
"configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_rank16.json"
.venv\Scripts\python scripts\train_sft_lora.py `
  --config $config `
  --local-low-vram `
  --max-train-rows 1000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

The candidate writes to its own `models` directory ending in `epochs2_rank16`. Do not overwrite the reference adapter or reduce context to work around memory pressure without recording that as a different experiment.

Results

Both runs use the established combined v1 + v2 development suite: 150 cases, with 81 ACT, 40 CLARIFY and 29 REFUSE expectations. The supplied evaluation command uses greedy decoding, a 256 new-token limit and the existing strict parser.

Metric	A: rank 8	B: rank 16
Full task successes	44/150 (0.293)	46/150 (0.307)
Routing macro-F1	0.514	0.450
Invalid outputs	2/150 (0.013)	7/150 (0.047)
Tool-call accuracy	0.600	0.547
Argument accuracy	0.771	0.785
Clarification wording accuracy	0.075	0.050

Metric	A: rank 8	B: rank 16
Refusal wording accuracy	0.172	0.103
Decision accuracy among known decisions	0.568	0.552
Decision coverage	0.987	0.953
Unavailable-tool hallucination rate	0.020	0.053
Average latency (ms)	1828	1980
Reported API cost (USD)	0.0000	0.0000

Ratios are rounded to three decimals. Wording scores are separate from routing: a correct clarification label can still miss the required question details. Local API cost zero does not mean training, hardware or electricity are free.

Routing Counts

Measurement	A: rank 8	B: rank 16
Correct ACT decisions	57/81	60/81
Correct CLARIFY decisions	11/40	11/40
Correct REFUSE decisions	16/29	8/29
Premature ACT on CLARIFY/REFUSE cases	31/69	42/69

Premature ACT here is the matrix count of productive decisions on boundary requests, not the potentially different count of the scorer's failure-reason label.

A: Rank-8 Decision Matrix

Rows are expected decisions; columns are recorded predictions. The diagonal measures correct routing, not full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	57	21	3	0
CLARIFY	20	11	9	0
REFUSE	11	0	16	2

B: Rank-16 Decision Matrix

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	60	14	2	5
CLARIFY	23	11	6	0

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
REFUSE	19	0	8	2

Invalid/unknown is the decision-matrix category. Invalid-output rate is computed separately from parser failures; the counts happen to match in these two runs.

Interpretation and Decision

The candidate completed two more tasks and slightly improved argument accuracy, but routing macro-F1 fell by 0.064. Correct refusals halved, premature ACT rose by eleven cases, and invalid outputs increased from two to seven. Correct clarification routing did not improve. Rank 16 is not a better balanced replacement.

Keep A, the rank-8 two-epoch SFT model, as the reference. This run does not support insufficient adapter capacity as the main problem. It does not prove capacity never matters, and a single-seed result cannot establish the cause of the regression or whether a two-task difference is reproducible.

The next test recommended after B was C: return to rank 8 / alpha 16 and lower only the learning rate from 2e-4 to 1e-4. That test has now been run; see Experiment C for its results. B remains a separate completed experiment, not part of a combined rank-and-learning-rate change.

Sources and Reproducibility

Artifact	Purpose
A evaluation JSON	Original reference results
B evaluation JSON	Rank-16 results, recorded at 2026-10-12T16:58:08.530056+00:00
B training config	Candidate settings and data identity
Comparison helper	Reads saved metrics and derives routing from recorded decisions

B canonical config SHA256: `a09ca9973edbc9127f7641cc5e10cb809baf18f13a3504c011350835fc0bd663`.

Numbers were checked against the saved result files using the comparison helper. No training, inference or result-file rewriting was performed to record this note. These repeatedly inspected suites are development comparisons, not an untouched final test. Exact base-model revision and complete generation/training metadata are not embedded in the result JSON, which limits retrospective verification.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.